



Campus Compass

A Human-Centred Campus Tour and Navigation
Web App for XJTLU Students and Visitors
Human-Centric Computing - CPT208

Group B3-5

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Introduction

Gap In Existing Solutions

- Existing solutions support navigation, but not onboarding. They help users move around campus, but do not fully support the process of getting started and settling in.
- Current solution types each address only part of the problem. **Commercial** maps provide live directions, official pages provide factual information, and **academic** systems improve route structure.
- Few solutions integrate onboarding and exploration for different campus user groups. Most do not combine route priorities, service resources, bilingual guidance, and confidence-building in one mobile experience.
- As a result, reaching a destination is not the same as feeling oriented. Users may arrive successfully, but still not know where to start, what matters most, or how to build familiarity and belonging on campus.

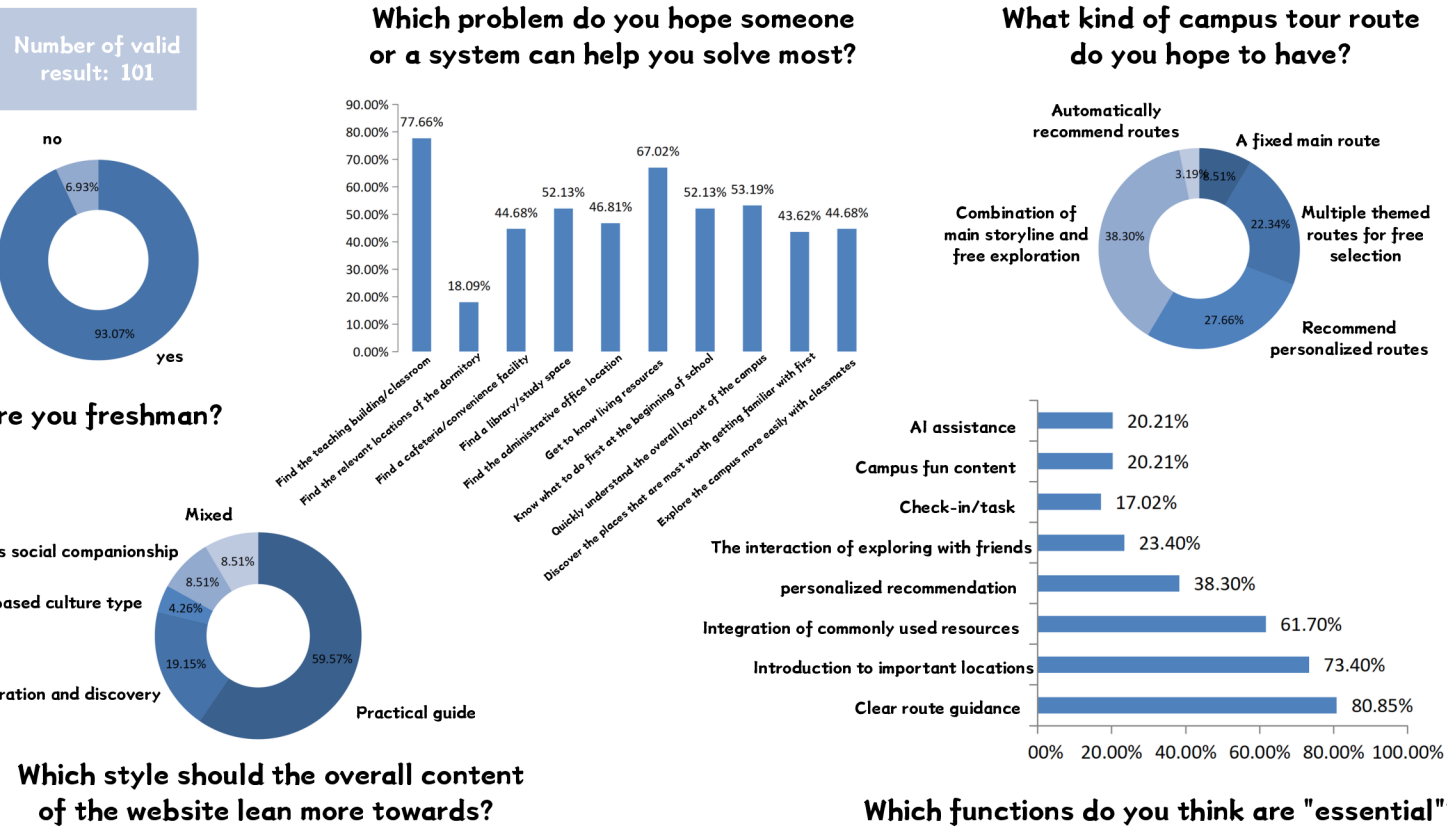
Motivation

- Campus Compass was developed to support campus onboarding beyond navigation. It helps new students and visitors understand where to begin, which places matter first, and how to move through campus with confidence, while also supporting broader needs such as exploration, service access, and practical campus tasks.
- Our findings revealed strong demand for clear route guidance, key place introductions, and integrated freshman resources. In response, we designed a human-centred system offering personalised routes, practical support, and a more guided, engaging campus experience.
- Therefore, the goal = improve navigation efficiency + strengthen users' confidence, familiarity, and sense of belonging on campus.

Requirement Gathering

Requirements Collected

The questionnaire grounded our requirements discovery by identifying the most urgent onboarding needs of first-year students. The findings highlighted clear route guidance, important-place introduction, and integrated campus resources, which were then translated into guided route entry, personalised recommendation, and practical campus-support features.



Design Requirements R1-R5

Mobile navigation must provide reliable, low-friction guidance through road-aligned paths, clear next steps, and visible progress.

Important campus destinations must be explained in context, not only pinned on a map, so users understand why each stop matters.

Different users need different route entry points and personalised routes.

High-priority campus support resources must be immediately accessible, especially for teaching, services, printing, and first-use tasks.

Primary: First-year students — mobile guidance, classrooms, services, first routes
Secondary: Visitors, parents, current students — tours, printing, services, light exploration



Prototype Design and Iteration

Aims and Objectives

To transform campus navigation into a guided and human-centred onboarding experience for XJTLU users:

- Reduce confusion and first-week anxiety for new students as they begin using the campus.
- Provide reliable route guidance to classrooms, service points, and other important campus destinations.
- Offer personalised and flexible route entry for different user needs and user groups.
- Combine navigation with practical campus support, including important-place introduction and access to key student resources.
- Encourage exploration and belonging through lightweight engaging features built on top of the practical core.

Design Alternatives

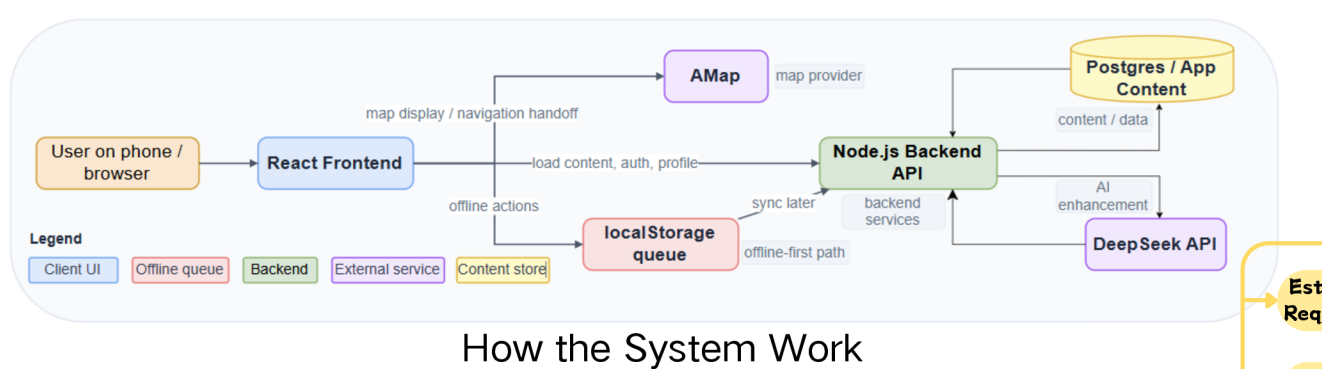
We did not jump from requirements straight to one polished interface. Instead, we compared route structures, questionnaire layouts, and visual directions before converging.

A. Route-structure alternatives

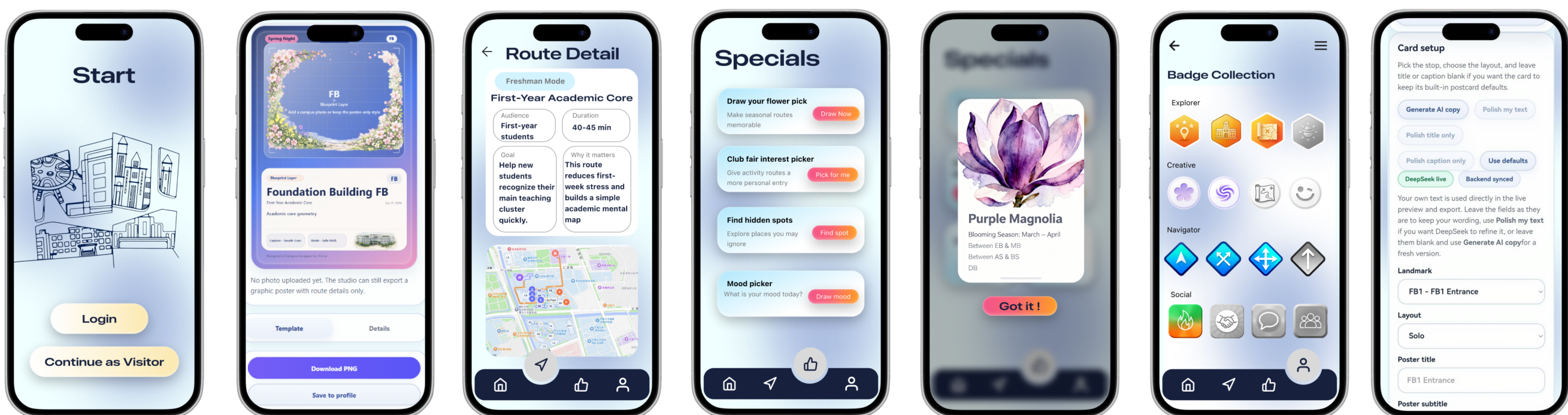
Alternative	Decision logic
Single freshman mainline	Too rigid for visitors and current students; weak fit with the survey preference for hybrid and personalised routes.
Multi-theme route library	Broader, but still weak for first-time users who need a clear starting suggestion.
Personalised recommender only	Strong for targeting, but too narrow for users who want to browse first.
Hybrid structure	Chosen. Recommendation + route library + optional exploratory layers best matched questionnaire findings.

B. Implemented questionnaire A/B alternatives

- Variant A: Guided form for steady step-by-step completion.
 - Variant B: Quick picks for faster chip-based answers with preset scenarios.
- It is human-centric through flexible, low-pressure route entry, and engaging through immediate, scenario-based participation.



Prototypes



Early Evaluation & Impact

Method Selection

Method	Why we used it	What it checks	Evidence in the alpha
Heuristic evaluation + cognitive walkthroughs	Fast formative review of first-use journeys before larger testing	Whether a newcomer can notice the right action, understand feedback, and progress with low confusion	Suitable for simulating first trips to classrooms, services, and memory-export flows
Task-based usability sessions	Matches the lecture focus on completion, duration, and observed friction	Can users finish the main tasks? How long does it take? Where do they hesitate?	User Test Mode records task, duration, rating, difficulty, blockers, comments, and exportable evidence
Route feedback capture	Supports continuous iteration after real route use	Perceived difficulty, confusing steps, and qualitative comments	Each route detail page includes an in-product feedback form
Admin evidence dashboard	Turns raw interactions into interpretable coursework evidence	Trends across submissions, A/B choices, completions, and user-test logs	Admin Insights summarises stored questionnaire, feedback, and evaluation records

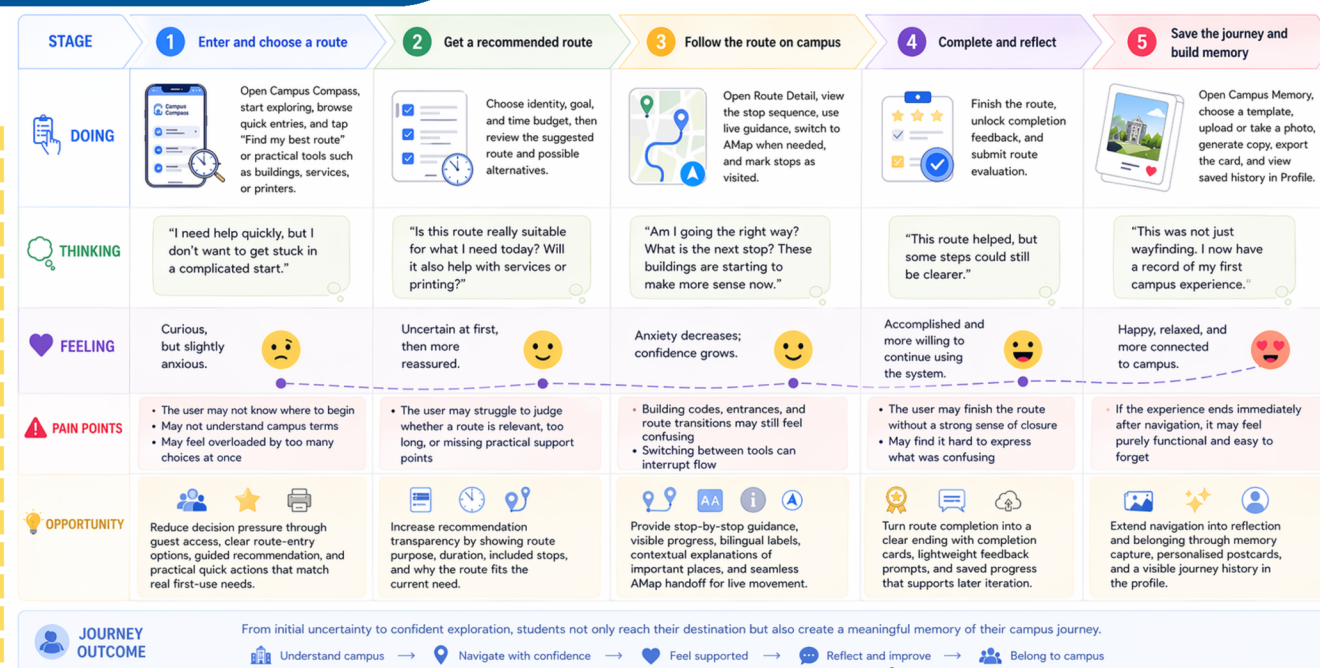
Task-based evaluation plan

- Task 1: Complete the recommendation questionnaire and understand the top route match.
- Task 2: Enter a route, follow the current leg, mark progress, and understand completion feedback.
- Task 3: Open Campus Memory, refine the generated content, and export a final card.

Impact

Campus onboarding is a human-centred challenge, not only a wayfinding challenge. This project matters because existing tools mainly help users move between locations, but do not fully support the early-campus decisions that matter most to newcomers, such as where to begin, which places matter first, and how to connect academic, service, and exploratory needs in one coherent journey. Campus Compass addresses this gap through a guided mobile experience that reduces decision pressure, supports practical first-use needs, and helps users build confidence, familiarity, and belonging on campus.

Journey Map



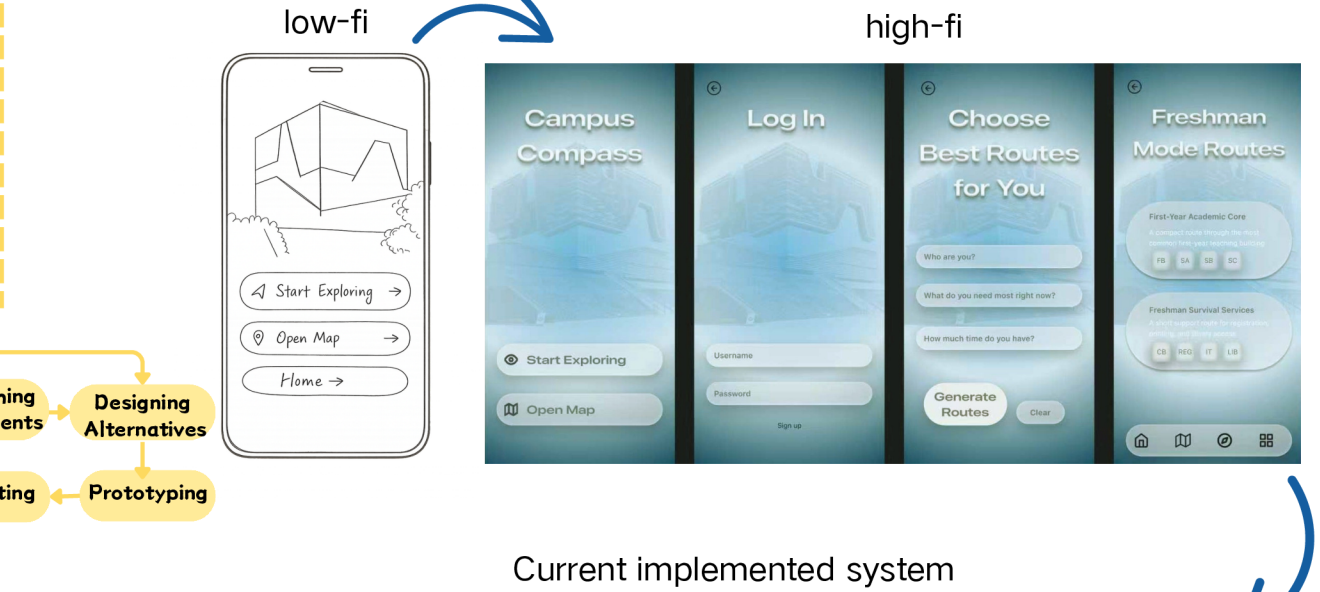
Novel Design Concept

- Campus Compass reframes campus touring as practical onboarding + playful belonging, not just a generic map task.
- It combines personalised route entry, practical campus support, and reflective memory features in one connected web experience.
- Users can start from different needs, but still move through a shared route-centred system.

Why It Is Engaging and Human-Centred

- Zero-barrier routing:** the interface does not assume prior knowledge of campus codes, landmarks, or jargon.
- Smart route entry:** recommendation, route families, and quick actions reduce decision pressure and help users begin from the right need.
- Practical-first support:** teaching buildings, services, printers, and route details are surfaced as first-class content rather than hidden utilities.
- Feedback and closure:** progress tracking, route completion, badges, and Campus Memory make navigation feel meaningful rather than merely consumed.
- Inclusive interaction:** bilingual guidance, mobile-first layout, and flexible route entry support different users, confidence levels, and campus contexts.

UI Alternatives



References

- [1] ChatGPT (GPT-5.3), version 5.3, accessed on 2026-04-10, available at <https://chat.openai.com>. Used for UI design prompt support.
- [2] Dreamina AI, version 1.0, accessed on 2026-04-10, available at <https://www.captur.com/dreamina>. Used for generating UI images and badge assets.
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- [1] T. Paetow, J. Wichmann, and M. Witzki. "Campus-Navigation-System Design for Universities: A Method Approach for Wismar Business School." In Human-Centred Intelligent Systems. A. Zimmermann, R. J. Howlett, L. C. Jain, and R. Schmidt. Eds. Singapore: Springer, 2021, pp. 3-12.
- [2] S. B. Olaleye, B. A. Adebiyi, A. Abdulaleem, F. C. Nwosu, A. O. Adeyemi, H. M. Ambi, and C. Omolayo. "Evaluating the usability and effectiveness of a special education campus navigation system for students with visual impairment." Journal of Intelligent Systems and Control, 2024, pp. 84-92. doi: 10.56578/jisc030202.
- [3] R. Tahir and J. Krogstad. "Impact of navigation aid and spatial ability skills on wayfinding performance and workload in indoor-outdoor campus navigation: Challenges and design." Applied Sciences, vol. 13, no. 17, Art. no. 9508, 2023. doi: 10.3390/app13179508.
- [4] Z. Li, G. Qi, W. Hu, X. Ma, and Q. Guo. "Application of augmented reality in campus navigation." in 2021 6th International Conference on Intelligent Computing and Signal Processing (ICSP), 2021, pp. 889-893. doi: 10.1109/ICSP51882.2021.9408770.

AI Disclosure

Conclusion

Campus Compass shows that campus navigation can be redesigned as a human-centred onboarding experience rather than a generic map tool. In its current alpha, it brings together requirement discovery, design alternatives, and implementation in one coherent system. Users can receive personalised route recommendations, follow practical mobile routes, access important buildings and services, track progress, and turn completed journeys into reflective Campus Memory postcards. Its main contribution is therefore not a single feature, but a route-centred design approach that helps users move from confusion to confidence while also supporting exploration and belonging.

Outcome

Limitation

The current alpha has a limited live evidence base, six core routes, and only early validation of its playful features, so broader usability testing is still required.

Future Work

Next steps include larger-scale user testing, expanded route coverage, stronger long-term gamification, and richer camera-based and user-generated exploration features.

